

## Article

# Forecasting the Behavior of Peruvian Coffee Export Prices in International Markets Using Econometric Models, 2010–2025

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## Abstract

Coffee export price volatility is a relevant problem for producing countries because it affects commercial planning, contract negotiation, producers' income stability, and the financial sustainability of the agro-export value chain. In economies that are highly dependent on primary commodities, abrupt fluctuations in international prices increase uncertainty and reduce the ability of economic agents to anticipate market behavior. In this context, the objective of this study was to forecast the behavior of the Peruvian coffee export price during 2025 by comparing econometric and time-series models. The research adopted a quantitative approach with a non-experimental, retrospective, and longitudinal design, using a monthly series for the 2010–2024 period. Seven specifications were estimated: linear model, quadratic model, Holt–Winters exponential smoothing, causal model, lagged model, ARIMA, and GARCH. The results showed that the GARCH (1,1) model achieved the best statistical performance, with the lowest Akaike Information Criterion, a Durbin–Watson statistic close to 2, an  $R^2$  higher than that of the alternative models, and no residual autocorrelation. Likewise, the significance of the ARCH and GARCH components confirmed the existence of volatility clustering in the series. The projections for 2025 show a fluctuating trajectory, although with a tendency to stabilize around values close to 10 from March onward. It is concluded that the GARCH (1,1) model is the most appropriate specification for forecasting the Peruvian coffee export price, as it provides a useful tool for export planning, risk management, and decision-making in a context of high uncertainty in the coffee market.

**Keywords:** price volatility; conditional heteroskedasticity; commodity markets; agricultural exports; risk management; time-series forecasting; international trade sustainability



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## 1. Introduction

Coffee is one of the most relevant agricultural commodities in international trade and provides income for more than 25 million households in developing countries, particularly in Africa, Latin America, and Asia, where its cultivation and commercialization have direct effects on rural employment and territorial sustainability [1]. Its global value chain is characterized by a high concentration of supply in producing economies, whereas demand is distributed across diversified international markets. In this context, Brazil and Vietnam lead global production and exports, while Colombia maintains a prominent position in the Arabica coffee segment, shaping supply dynamics and the transmission of shocks to international prices and global supply chains [2]. Coffee also remains one of the most traded

and consumed agricultural products worldwide, with sustained growth in demand over recent decades [3]. Therefore, analyzing and anticipating its price trajectory is strategically important for producers, exporters, and public policymakers [4].

The specialized literature recognizes export diversification as a key mechanism for reducing the vulnerability of economies exposed to fluctuations in coffee and other commodity prices. Empirical evidence indicates that persistent dependence on primary products may result in lower long-term returns and high income volatility, thereby affecting both macroeconomic stability and the sustainability of foreign trade. Within this framework, several authors have emphasized the need to promote diversification strategies and stabilization policies to reduce the fragility associated with income derived from coffee and other products [5,6]. Consequently, coffee should be understood not only as an economically relevant export product, but also as a structural component in shaping the trade specialization pattern of multiple economies.

The strategic importance of coffee is also reflected in the scale of its production. According to the USDA Foreign Agricultural Service, global production exceeded 171.4 million 60 kg bags in the 2023/2024 season, representing a 5.82% increase compared with the previous cycle [7]. In this scenario, Brazil ranks first worldwide and accounts for approximately 35% of production, with nearly 63 million bags annually [8]. Colombia, in turn, maintains a relevant position in the international Arabica coffee market despite the adverse effects its coffee sector has faced as a result of climate change [9]. In the case of Peru, the country remains an important producer of Arabica coffee, with output close to 4.2 million 60 kg bags and product differentiation associated with organic coffee. However, this competitive position faces constraints related to climatic factors, phytosanitary pressure, labor costs, and new regulatory requirements in destination markets, particularly in the European Union, where traceability and due diligence requirements have been reinforced under anti-deforestation regulations [10].

According to estimates from the USDA Foreign Agricultural Service, world coffee production for the 2024/25 season was forecast at 176.2 million 60 kg bags, while global exports were projected at 123.1 million bags and consumption at 170.6 million bags. In the case of Peru, coffee production for the 2025/26 season was forecast at 4.2 million 60 kg bags, representing an 8% increase over the previous year. These figures confirm that Peru remains a relevant Arabica coffee producer, but they also show that its export performance depends on a highly competitive and volatile international market. Therefore, forecasting export prices is not only a technical exercise, but also a strategic input for commercial planning, risk management, and the economic sustainability of the coffee value chain [10].

One of the factors that most strongly contributes to uncertainty in the coffee sector is climate change. Evidence shows that this phenomenon is altering the agroclimatic suitability of coffee cultivation, intensifying extreme events, and increasing the incidence of pests and diseases, with direct effects on yield, quality, and the economic viability of production systems [11]. At the international level, rising temperatures have been shown to favor the spread of the coffee berry borer and to increase global production risk [12]. Likewise, climate projections anticipate a significant reduction in land suitable for coffee production by the middle of the century, with important implications for producing regions in the Americas, Asia, and Africa [13]. In Peru, these risks are especially significant because of the vulnerability of coffee production to coffee leaf rust, a disease that has caused yield losses and changes in producers' commercial decisions, including those of organic cooperatives [14]. This is compounded by the genetic diversity and emergence of new races of *Hemileia vastatrix*, which increase the complexity of sanitary management and reinforce the need for adaptation strategies based on scientific evidence [15].

The effects of these climatic and sanitary shocks are transmitted to the market through changes in export availability, contract compliance, and price formation. In the Peruvian case, Ref. [16] quantified income losses affecting 110,227 producers and identified a reduction of between 8% and 13% in expected income per hectare of coffee due to climate change. Likewise, Ref. [17] documented substantial economic losses in coffee farms in northeastern Peru, where climatic factors, particularly the El Niño events of 2012 and 2016, significantly reduced production. These pressures have been reinforced by recent regulatory changes, such as the European Union Deforestation Regulation and the Corporate Sustainability Due Diligence Directive, as well as by market dynamics associated with rising green coffee prices, which contributed to the increase in coffee prices in 2024 [18]. In agricultural commodity markets, such episodes amplify temporary imbalances between supply and demand and increase uncertainty among economic agents, especially when profitability depends on both yield and selling price [19].

In this environment, price forecasting acquires strategic value for commercial decision-making. Producers and exporters must decide not only how much to produce, but also when to sell, how to negotiate contracts, and which strategies to adopt to reduce exposure to market risk. Therefore, having reliable forecasting tools can contribute to better commercial planning and strengthen the resilience of the coffee value chain.

In particular, the European Union Deforestation Regulation requires operators and traders to demonstrate that products placed on the EU market are deforestation-free, thereby increasing traceability and due diligence requirements for commodities such as coffee. Likewise, Directive (EU) 2024/1760 on corporate sustainability due diligence strengthens corporate responsibility regarding adverse impacts on human rights and the environment throughout value chains. These regulatory changes are especially relevant for Peruvian coffee exporters, since compliance with traceability, sustainability, and documentation requirements may affect market access, transaction costs, and export planning [20–22].

Despite the economic relevance of coffee and methodological advances in forecasting techniques, comparative analyses of econometric models specifically applied to the Peruvian coffee export price remain limited. Most previous studies have focused on international benchmark prices, production yields, or general methodological applications, whereas empirical evidence specifically addressing the Peruvian export price remains scarce. This gap is relevant because the export price does not depend exclusively on international market dynamics, but also on factors related to quality, product composition, commercial structure, logistics, and regulatory requirements in destination markets. Moreover, coffee continues to play an important role in Peru's agro-export performance, although it faces constraints derived from price volatility, international competition, and logistics costs [23,24].

Within this context, the present study offers an applied contribution focused on forecasting the export price of Peruvian coffee. To this end, it analyzes a monthly series for the 2010–2024 period, compares seven econometric and time-series specifications, linear, quadratic, exponential smoothing, causal, lagged, and ARIMA models, and generates projections for 2025. The aim is to identify the model with the best predictive performance and to provide a useful tool for decision-making in sales, contract negotiation, logistics scheduling, and risk management. Accordingly, the general objective of the study is to forecast the behavior of the Peruvian coffee export price in international markets for 2025 using econometric models. More specifically, the study seeks to estimate the coffee export price through different econometric models and to compare their predictive accuracy in order to determine the most effective specification.

Within this framework, the present study aims to forecast the behavior of the Peruvian coffee export price in international markets for 2025 using econometric models. To this end,

different model specifications are estimated and their predictive accuracy is compared in order to identify the most appropriate model for forecasting the Peruvian coffee export price and to provide a useful tool for commercial decision-making.

The case of Peruvian coffee was selected because it combines economic relevance, territorial importance, and high exposure to external shocks [25]. Coffee production in Peru is mainly associated with small and medium-sized producers located in rural and environmentally sensitive areas, where export income contributes to household livelihoods and local economic activity. At the same time, the sector faces climate risks, phytosanitary pressures, logistics constraints, and increasing sustainability requirements in destination markets. Therefore, price forecasting is directly linked to sustainability: from a financial perspective, it supports income stability and risk management; from a territorial perspective, it contributes to planning in coffee-growing areas; and from a commercial perspective, it improves exporters' ability to anticipate market fluctuations and comply with international requirements [26–28].

### 1.1. Literature Review

#### 1.1.1. Theoretical Foundations of Supply, Demand, and Price Formation

This study is grounded in supply and demand theory as a general framework for explaining price formation and its adjustments in response to shocks. From this perspective, price represents the monetary expression of consumers' willingness to pay, whereas demand reflects the quantity of goods that consumers are willing to purchase at different price levels, maintaining an inverse relationship between the two variables [29]. Supply, in turn, describes the quantity that producers are willing to place on the market according to expected prices, costs, and available technology. Equilibrium is reached when both forces are equal, whereas deviations generate excess supply or excess demand [30].

In the case of agricultural commodities, this conceptual foundation is complemented by evidence on price transmission. Domestic or export prices tend to adjust, although with frictions and possible asymmetries, to changes in international prices, especially in liberalized markets and extensive global supply chains [31]. Consequently, the analysis of the coffee export price requires consideration of both market fundamentals and the interaction between external and internal shocks.

Price transmission theory examines how price changes at one market level are transferred to other stages of the value chain. Empirical evidence in agricultural sectors shows that asymmetric transmission is common. In this regard, Ref. [32] found that such asymmetry is more likely in sectors with fragmented production structures, whereas it tends to decline when retail regulations limit entry barriers. Likewise, Ref. [33] shows that the presence of intermediaries with monopsony power may intensify these asymmetries, generating stronger transmission when international prices decline than when they increase. Taken together, these findings indicate that market structure significantly influences price transmission dynamics.

Market integration theory argues that prices across different markets tend to converge when there is an efficient flow of information and arbitrage. Several studies have confirmed this mechanism in the international coffee market. Ref. [34] found that coffee markets in Brazil exhibit spatial integration, thereby favoring arbitrage and the operation of the Law of One Price. Complementarily, Ref. [35] shows that the main Arabica coffee-producing countries, Brazil, Colombia, Mexico, Guatemala, Peru, and Honduras, are integrated with international prices, with Brazil standing out as the country with the highest degree of integration due to its share in global exports.

The literature on commodity price cycles identifies several macroeconomic and structural factors that influence international price dynamics. Among the most relevant deter-

minants are industrial growth, monetary expansion, dollar depreciation, and inventory levels [36]. Likewise, variables such as exchange rates, real interest rates, and international liquidity also play an important role [37]. Other studies highlight the influence of climatic factors, inflation in the United States, the growth of emerging markets, and inventory availability [38]. More recently, geopolitical shocks such as conflicts or pandemics have increased price volatility [39], while structural factors such as agricultural productivity and energy costs continue to exert persistent pressure on prices [34].

#### 1.1.2. Time-Series Forecasting: Benchmark Models and Evaluation Criteria

Methodological literature recognizes ARIMA/SARIMA models and exponential smoothing as benchmark approaches for univariate forecasting when trend, seasonality, and temporal dependence are present [40]. These models formalize widely used procedures and incorporate error metrics such as MAE, RMSE, and MAPE to compare specifications and select the one with the best predictive performance. Therefore, they constitute robust baseline approaches in time-series studies, especially when the goal is to generate short-term forecasts under replicable criteria [41].

In coffee markets, shocks have also been shown to exhibit high persistence, increasing the complexity of modeling and reinforcing the need to compare alternative specifications in order to assess how quickly changes in the price path dissipate [42].

#### 1.1.3. Empirical Evidence on Coffee Price Forecasting: Linear and Nonlinear Approaches

In the specific field of coffee price forecasting, Ref. [43] proposes a methodology that compares classical approaches, such as AR, ARIMA, and exponential smoothing, with neural networks, particularly MLP and ELM. Their results show that nonlinear methods may outperform traditional models when time-series preprocessing and the identification of complex patterns are properly implemented. This finding suggests that predictive accuracy depends not only on the type of model, but also on the structure of the analyzed series.

Additionally, when the focus is not limited to the mean level of the price but also includes its volatility, the literature recommends ARIMA–GARCH-type approaches. In this regard, Ref. [44] applies an econometric framework to forecast coffee prices while incorporating volatility dynamics, which confirms the relevance of accounting for heteroskedasticity in commodity markets.

#### 1.1.4. Hybrid Approaches and Foreign Trade Forecasting: Implications for Export Prices

For foreign trade variables such as exports or export prices, where linear components and nonlinear patterns often coexist, improvements have been reported through hybrid approaches. Ref. [45] integrates ARIMA and LSTM in a hybrid model for export forecasting and show that the combination reduces forecasting errors compared with the isolated use of each technique. This result supports the convenience of comparing different model families when the series exhibits structural complexity.

In the specific case of export prices, Ref. [46] compares the BetaSutte model with ARIMA in forecasting Indonesia's export price and highlight the importance of conducting out-of-sample validation using homogeneous metrics. Similarly, Ref. [47] demonstrates the applicability of ARIMA to international trade variables, specifically in forecasting cashew export volumes in Vietnam.

Overall, the literature agrees that the selection of a forecasting model should be based on empirical comparisons, error metrics, and consistent validation procedures. In line with this approach, the present study compares seven specifications—linear, quadratic, exponential smoothing, causal, lagged, and ARIMA—in order to identify the model with the highest predictive accuracy for forecasting the Peruvian coffee export price.

A critical reading of the literature reveals three relevant conclusions for this study. First, traditional time-series models, such as ARIMA and exponential smoothing, remain useful benchmarks because they provide interpretable and replicable forecasts. Second, nonlinear and hybrid approaches may improve predictive accuracy when the series contains complex patterns that cannot be captured by linear specifications. Third, when the research problem involves price volatility rather than only the behavior of the average price, ARCH/GARCH-type models become particularly relevant because they explicitly model conditional variance. Therefore, the methodological contribution of this study is not based on assuming the superiority of any model *a priori*, but on comparing alternative specifications using the same monthly series of Peruvian coffee export prices. This comparative design allows the model selection process to be guided by empirical performance, diagnostic criteria, and the specific behavior of the analyzed series.

## 2. Materials and Methods

This study adopted a quantitative approach with a non-experimental and retrospective longitudinal design aimed at forecasting the export price of Peruvian coffee [48,49]. From a comparative perspective, seven econometric and time-series specifications were estimated in order to identify the model with the best predictive performance for the 2025 horizon [50].

The unit of analysis consisted of a monthly time series of the Peruvian coffee export price corresponding to national tariff code 0901.11.90.00 for the period 2010–2024 (Customs Tariff 2022) [51,52]. The dependent variable ( $P_t$ ) represents the export price expressed in U.S. dollars. For the causal and lagged models, FOB value and net export weight were included as covariates, both in contemporaneous form and with a one-period lag ( $t$  and  $t - 1$ , respectively), in order to assess whether these variables provide additional predictive power relative to univariate approaches.

The dataset was cleaned using inclusion and exclusion criteria to retain only records corresponding to green coffee, excluding non-relevant products and by-products. The data were then aggregated to a monthly frequency for estimation and forecasting.

For the econometric analysis, seven forecasting models were estimated. First, a linear model with deterministic trend was considered, specified as  $P_t = \alpha + \beta \cdot t + \varepsilon_t$  [53], together with a quadratic model defined as  $P_t = \alpha + \beta_1 \cdot t + \beta_2 \cdot t^2 + \varepsilon_t$  [54]. In addition, an additive Holt–Winters exponential smoothing model with monthly seasonality ( $m = 12$  m) was applied, decomposing the series into level, trend, and seasonal components through smoothing parameters ( $\alpha, \beta, \gamma$ ) [55,56].

Furthermore, a benchmark causal model was estimated, specified as  $P_t = \alpha + \gamma_1 \cdot \text{FOB}_t + \gamma_2 \cdot \text{Peso}_t + \varepsilon_t$ , along with a lagged model defined as  $P_t = \alpha + \delta_1 \cdot \text{FOB}_{t-1} + \delta_2 \cdot \text{Peso}_{t-1} + \varepsilon_t$  [57]. Finally, an ARIMA model was estimated following the Box–Jenkins methodology. Different ARIMA ( $p, d, q$ ) specifications were evaluated, and the model with the best fit and residual diagnostics according to information criteria was selected. The final specification corresponds to an ARIMA (1,0,0) model, expressed as  $P_t = c + \phi_1 \cdot P_{t-1} + \varepsilon_t$  [58,59].

The analytical procedure was conducted in two complementary stages. First, the models were estimated and their statistical fit was assessed. Second, their predictive performance was compared using standard metrics, including Mean Squared Error (MSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), Theil's inequality coefficient, and the coefficient of determination ( $R^2$ ). This procedure allowed the identification of the most robust model for forecasting the behavior of the Peruvian coffee export price for the 2025 horizon [60,61].

All estimations, diagnostics, and forecasts were performed using EViews 9.0 software, consistent with the outputs reported in the tables and figures of the manuscript. The data

correspond to secondary foreign trade records aggregated at a monthly frequency. Since the study did not involve human subjects or personal data, no specific ethical approval was required [62].

The empirical evidence summarized in Table 1 indicates that the performance of forecasting models applied to agricultural commodity prices, particularly coffee, varies depending on the structure of the time series and the methodological approach employed. In general, traditional econometric models, such as ARIMA and its extensions, show consistent performance in the short term, especially when the series exhibits stable temporal dependence patterns. However, several studies highlight that nonlinear approaches, including neural networks and hybrid models (e.g., ARIMA–LSTM), tend to improve predictive accuracy by capturing complex and nonlinear relationships. In addition, the incorporation of volatility models, such as ARIMA–GARCH, is particularly relevant in contexts characterized by high price variability. Overall, these findings suggest that there is no universally superior model; rather, the optimal choice depends on the specific characteristics of the data and the forecasting horizon, thereby supporting the comparative approach adopted in this study.

**Table 1.** Empirical evidence on forecasting models applied to coffee and agricultural commodity prices.

| Author                   | Method                                   | Data                                                                                              | Main Finding                                                                                                                                                                                                                                                                                |
|--------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Larre & Auza (2020) [63] | ARIMA-GARCH                              | Historical series of Merval index returns                                                         | The ARIMA-GARCH model captures coffee price volatility and improves the accuracy of forecasts                                                                                                                                                                                               |
| Deina et al. (2022) [64] | Extreme Learning Machine (ELM) vs. ARIMA | Price data for two types of coffee                                                                | Neural networks, including ELM, outperform linear models such as ARIMA in predicting coffee prices.                                                                                                                                                                                         |
| Gasmi et al. (2024) [65] | ARIMA–LSTM hybrid model                  | Annual precipitation                                                                              | The ARIMA–LSTM hybrid model significantly reduces prediction errors compared to individual ARIMA and LSTM models by leveraging the advantages of both linear and nonlinear techniques.                                                                                                      |
| David et al. (2017) [66] | Fractional integration models            | Time series of prices for four major agricultural commodities: soybeans, corn, coffee, and sugar. | ARFIMA and ARIMA models with coffee price data explicitly demonstrated that “ARFIMA outperforms in forecasting future prices.”                                                                                                                                                              |
| Ahmar & Val (2025) [46]  | BetaSutte vs. ARIMA                      | Monthly export values                                                                             | They compared these models using Indonesian export prices and found that BetaSutte achieved superior out-of-sample accuracy. It is worth noting that ARIMA demonstrated better in-sample performance, suggesting that BetaSutte’s advantage arises specifically in out-of-sample forecasts. |
| Ahmar & Val (2025) [46]  | ARIMA                                    | Historical prices                                                                                 | The ARIMA model is effective for short-term forecasting of agricultural commodity prices                                                                                                                                                                                                    |

Note. Authors’ own elaboration based on a review of the scientific literature. The table summarizes empirical studies comparing econometric, machine learning, and hybrid approaches for forecasting agricultural commodity prices, particularly coffee, highlighting differences in predictive performance.

### 3. Results

To analyze the behavior of the Peruvian coffee export price in international markets, seven forecasting models were estimated: linear, quadratic, exponential smoothing, causal, lagged, and ARIMA. All estimations were performed using EViews 9.0 software.

### 3.1. Estimating the Export Price of Coffee Using Econometric Models

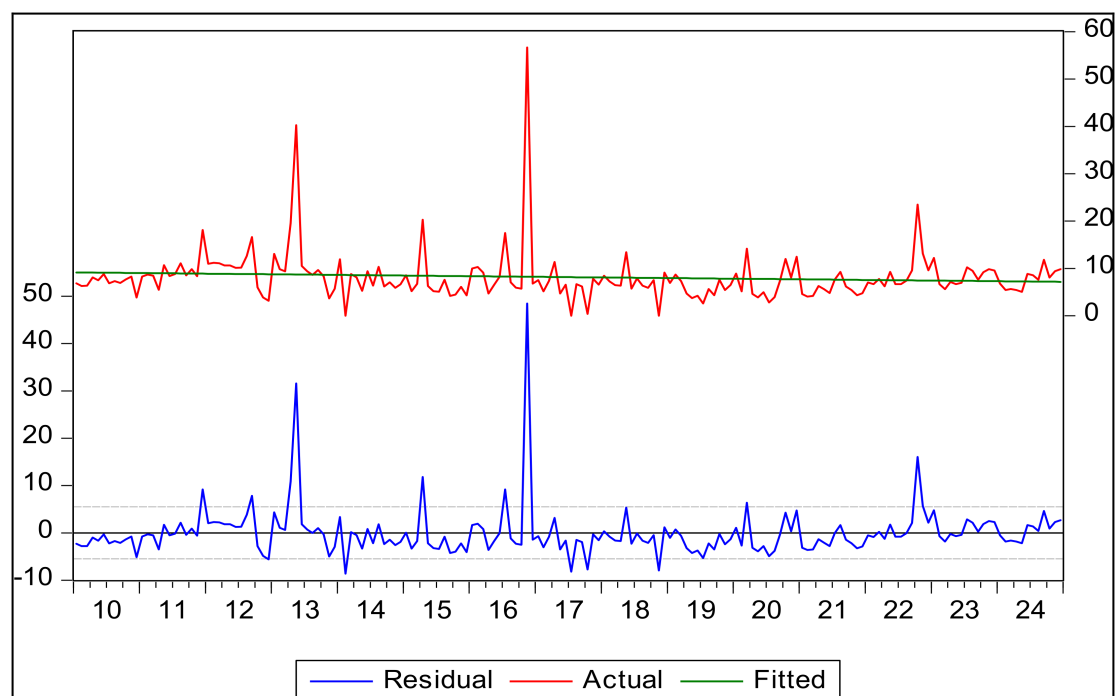
As an initial approach, a linear forecasting model was estimated for the Peruvian coffee export price. The results show that the trend coefficient was negative and statistically non-significant ( $\beta = -0.011$ ;  $p = 0.162$ ), indicating that the series does not exhibit a significant linear trend over the analyzed period. Therefore, the linear specification does not adequately explain the evolution of the coffee export price.

In addition, the model displayed very limited explanatory power, with an  $R^2$  of 0.01 and an adjusted  $R^2$  of 0.01, suggesting that only 1% of the observed variation in the export price is explained by the included linear trend. Likewise, the Akaike Information Criterion was 6.255, while the Durbin–Watson statistic reached 1.755, suggesting the possible presence of residual autocorrelation. Overall, these findings indicate that the linear model provides only a preliminary approximation and does not offer a sufficiently robust fit to describe the dynamics of the Peruvian coffee export price, as shown in Table 2 and Figures 1 and 2.

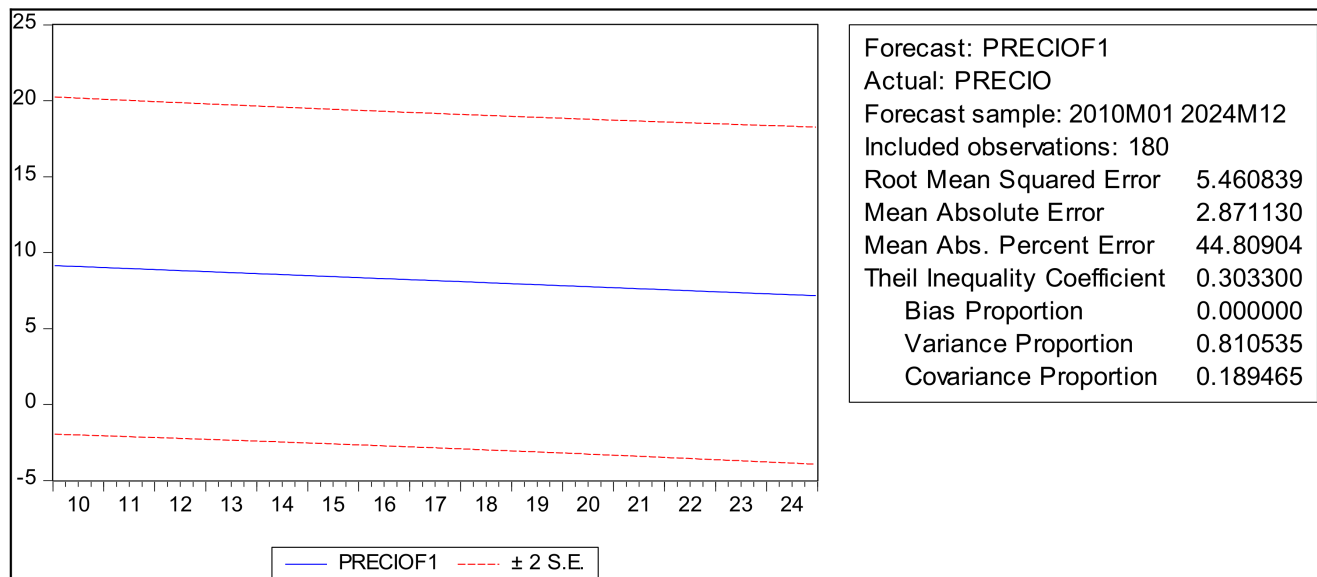
**Table 2.** Linear Forecast Estimate.

| Variable                     | Coefficient | Standard Error               | t-Statistic | Prob. |
|------------------------------|-------------|------------------------------|-------------|-------|
| C                            | 9.144       | 0.815                        | 11.216      | 0.000 |
| @TREND                       | −0.011      | 0.008                        | −1.404      | 0.162 |
| R-squared                    | 0.01        | Mean dependent variable      |             | 8.154 |
| Adjusted R-squared           | 0.01        | S.D. of dependent variable   |             | 5.506 |
| Standard error of regression | 5.49        | Akaike information criterion |             | 6.255 |
| Sum of squared residuals     | 5367.74     | Schwarz criterion            |             | 6.291 |
| Log-likelihood               | −560.98     | Hannan–Quinn criterion       |             | 6.270 |
| F-statistic                  | 1.97        | Durbin–Watson statistic      |             | 1.755 |
| Prob (F-statistic)           | 0.16        |                              |             |       |

Note: Prepared using Eviews 9.0 software.



**Figure 1.** Residuals from the linear forecast. Note: Prepared using Eviews 9.0 software.



**Figure 2.** Linear model forecast. Note: Prepared using Eviews 9.0 software.

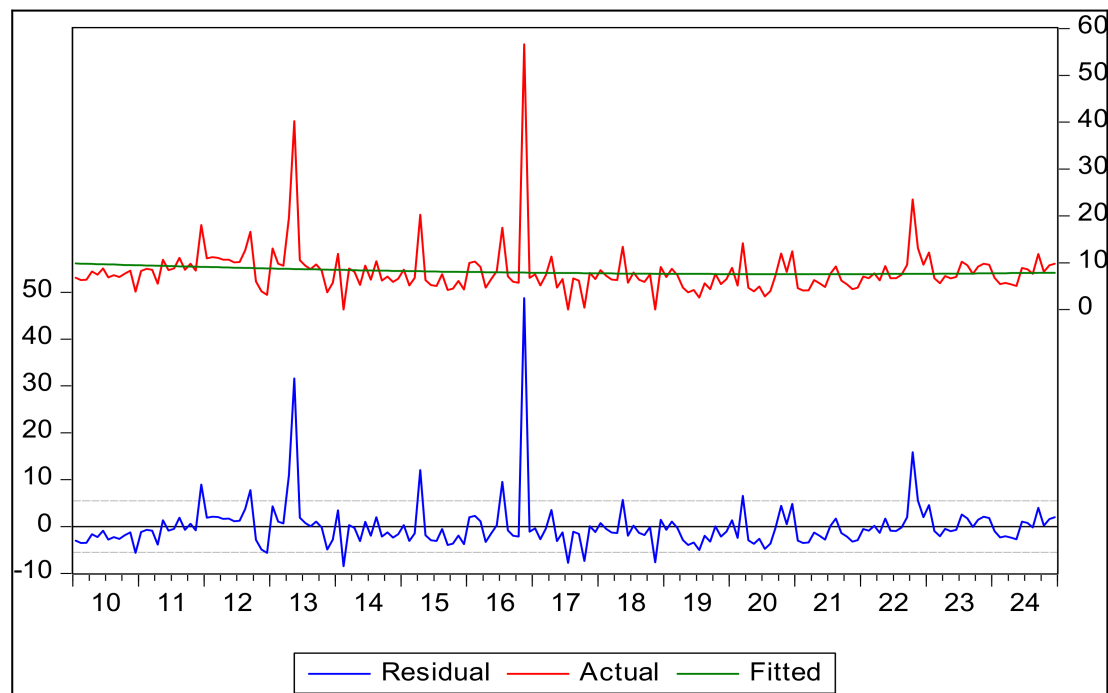
Subsequently, a quadratic forecasting model was estimated for the Peruvian coffee export price in order to assess the possible presence of a nonlinear trajectory in the series. The results show that neither the linear trend term tendency (@TREND;  $\beta = -0.035$ ;  $p = 0.270$ ) nor the quadratic term (@TREND<sup>2</sup>;  $\beta = 0.000$ ;  $p = 0.440$ ) was statistically significant. This indicates that the behavior of the export price cannot be adequately explained by a quadratic trend over the analyzed period.

Likewise, the model displayed very limited explanatory power, with an  $R^2$  of 0.014 and an adjusted  $R^2$  of 0.003, indicating only a marginal contribution of this specification to explaining the observed variation in export prices. Similarly, the overall model was not statistically significant ( $F = 1.288$ ;  $p = 0.278$ ), confirming the weak statistical relevance of the quadratic fit. In comparative terms, the Akaike Information Criterion was 6.26, while the Durbin–Watson statistic reached 1.76, suggesting the possible presence of residual autocorrelation. Overall, these findings indicate that the quadratic model does not substantially improve the explanation of the behavior of the Peruvian coffee export price, as presented in Table 3 and Figures 3 and 4.

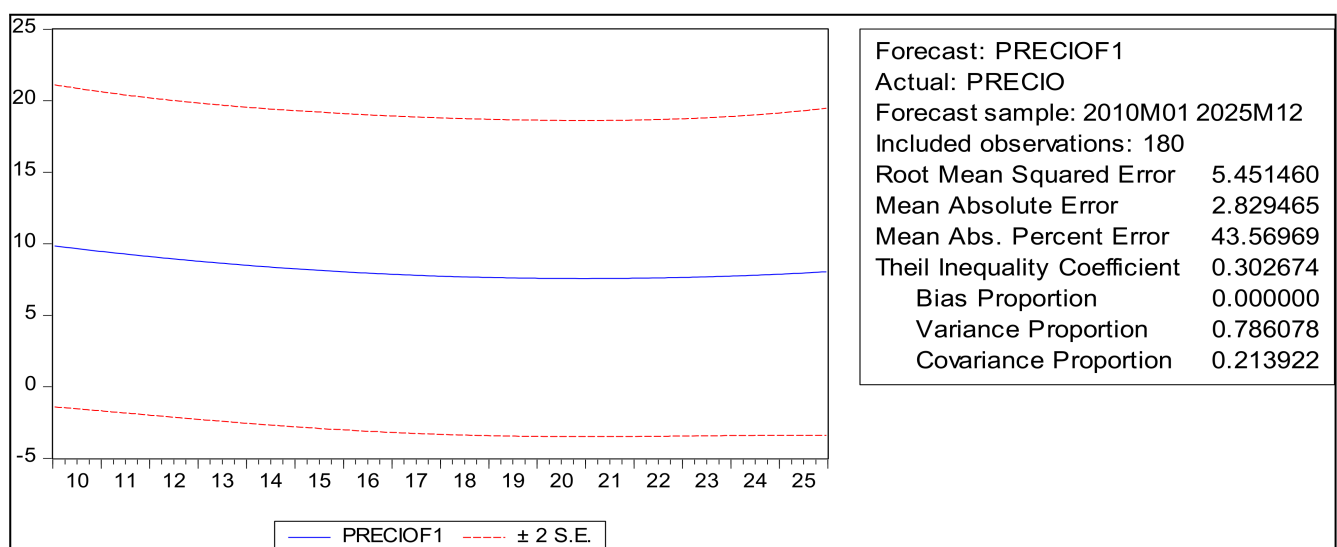
**Table 3.** Estimation of the quadratic forecast.

| Variable                     | Coefficient | Std. Error                   | t-Statistic | Prob. |
|------------------------------|-------------|------------------------------|-------------|-------|
| C                            | 9.847       | 1.216                        | 8.100       | 0.00  |
| @TREND                       | −0.035      | 0.031                        | −1.108      | 0.27  |
| @TREND <sup>2</sup>          | 0.000       | 0.000                        | 0.781       | 0.44  |
| R-squared                    | 0.014       | Mean dependent variable      |             | 8.15  |
| Adjusted R-squared           | 0.003       | S.D. of dependent variable   |             | 5.51  |
| Standard error of regression | 5.497       | Akaike information criterion |             | 6.26  |
| Sum of squared residuals     | 5349.314    | Schwarz criterion            |             | 6.32  |
| Log-likelihood               | −560.668    | Hannan–Quinn criterion       |             | 6.28  |
| F-statistic                  | 1.288       | Durbin–Watson statistic      |             | 1.76  |
| Prob (F-statistic)           | 0.278       |                              |             |       |

Note: Generated using Eviews 9.0.



**Figure 3.** Residuals from the quadratic forecast. Note: Prepared using Eviews 9.0 software.



**Figure 4.** Forecast of the quadratic model. Note: Prepared using Eviews 9.0.

Subsequently, a Holt–Winters exponential smoothing model with additive specification was estimated in order to assess the contribution of the level, trend, and seasonal components to the coffee export price series. The results show that the level parameter ( $\alpha = 0.05$ ) provides a moderate adjustment, whereas the trend ( $\beta = 0$ ) and seasonal ( $\gamma = 0$ ) components do not show relevant contributions within the model structure.

In terms of fit, the model reported a Root Mean Squared Error (RMSE) of 5.449346 and a Sum of Squared Residuals of 5345.168, reflecting a limited ability to reduce forecast error variability. In addition, the end-of-period estimates show a mean level of 8.172327 and a nearly null trend (0.003864), suggesting a relatively stable series dynamic.

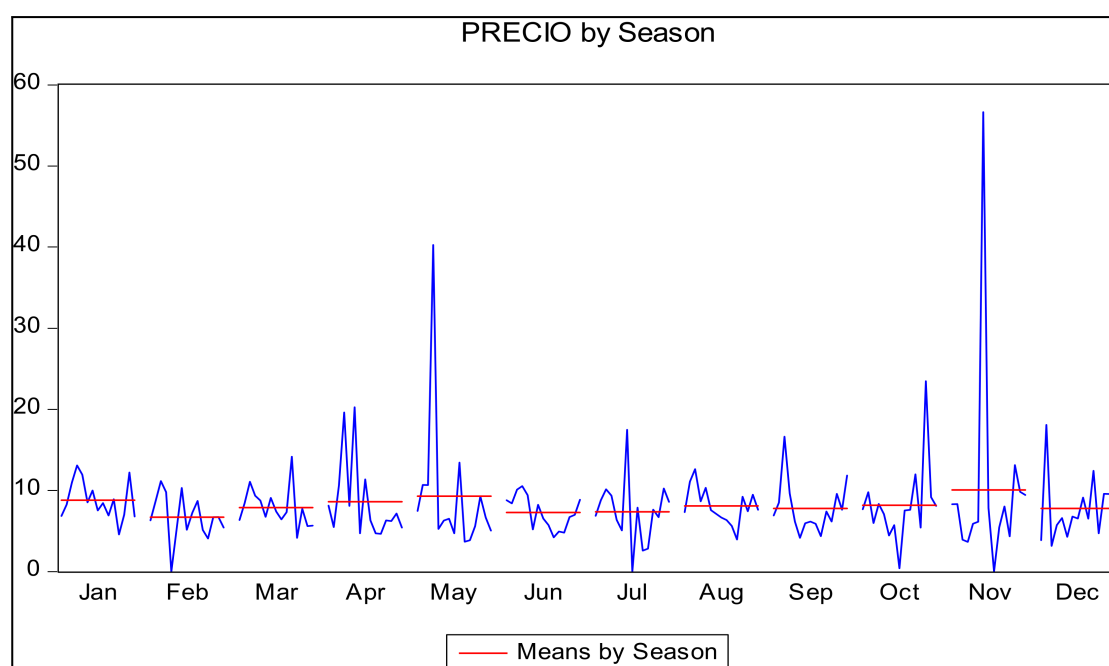
Regarding the seasonal component, although differentiated monthly variations are observed, these do not constitute a robust seasonal pattern within the model, given that the gamma parameter is equal to zero. Overall, these findings suggest that the additive Holt–

Winters specification does not consistently capture the dynamics of the coffee export price, particularly with respect to trend and seasonality, as presented in Table 4 and Figure 5.

**Table 4.** Estimation of the exponential smoothing forecast.

|                          |            |                   |
|--------------------------|------------|-------------------|
| Parameters:              | Alpha      | 0.05              |
|                          | Beta       | 0                 |
|                          | Gamma      | 0                 |
| Sum of Squared Residuals |            | 5345.168          |
| Root Mean Square Error   |            | 5.449346          |
| End-of-Period Levels:    | Mean       | 8.172327          |
|                          | Trend      | 0.003864          |
|                          | Seasonals: | 2024M01 0.665919  |
|                          |            | 2024Q2 −1.433945  |
|                          |            | 2024M03 −0.252476 |
|                          |            | 2024M04 0.460327  |
|                          |            | 2024M05 1.145796  |
|                          |            | 2024M06 −0.858068 |
|                          |            | 2024M07 −0.790599 |
|                          |            | 2024M08 −0.087796 |
|                          |            | 2024M09 −0.364327 |
|                          |            | 2024M10 0.014476  |
|                          |            | 2024M11 1.889945  |
|                          |            | 2024M12 −0.389252 |

Note: Prepared using Eviews 9.0 software.



**Figure 5.** Residuals from the exponential smoothing model. Note: Prepared using Eviews 9.0 software.

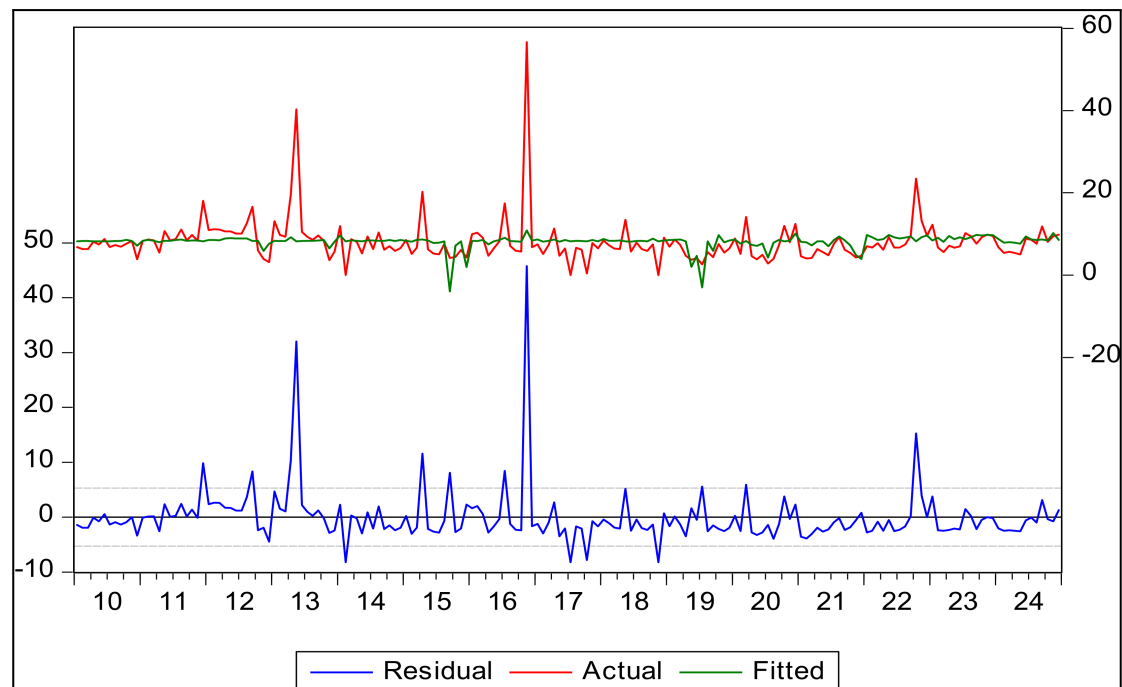
In a subsequent stage, a causal model was estimated to explain the behavior of the Peruvian coffee export price based on FOB value and net weight. The results show that both variables exert statistically significant effects on the export price. Specifically, FOB value displayed a positive and significant relationship ( $\beta = 3.67 \times 10^{-5}$ ;  $p = 0.0143$ ), whereas net weight showed a negative and also significant effect ( $\beta = -0.000214$ ;  $p = 0.0011$ ). These findings suggest that, within the estimated specification, a higher FOB value is associated with increases in the export price, while larger net export volumes tend to be associated with a reduction in price.

At the overall level, the model was statistically significant ( $F = 8.823413$ ;  $p = 0.000222$ ), confirming that the included variables jointly contribute to explaining the evolution of the coffee export price. However, its explanatory power remains limited, with an  $R^2$  of 0.090661 and an adjusted  $R^2$  of 0.080386, indicating that approximately 9.1% of the observed variation in price is explained by the model. In terms of fit, the Akaike Information Criterion was 6.182389, while the Durbin–Watson statistic reached 1.821858, suggesting the possible presence of mild residual autocorrelation, although less pronounced than in the previously estimated models. Overall, these results indicate that the causal model improves explanatory capacity relative to the linear and quadratic specifications; nevertheless, it still has limitations in broadly capturing the dynamics of the Peruvian coffee export price, as shown in Table 5 and Figures 6 and 7.

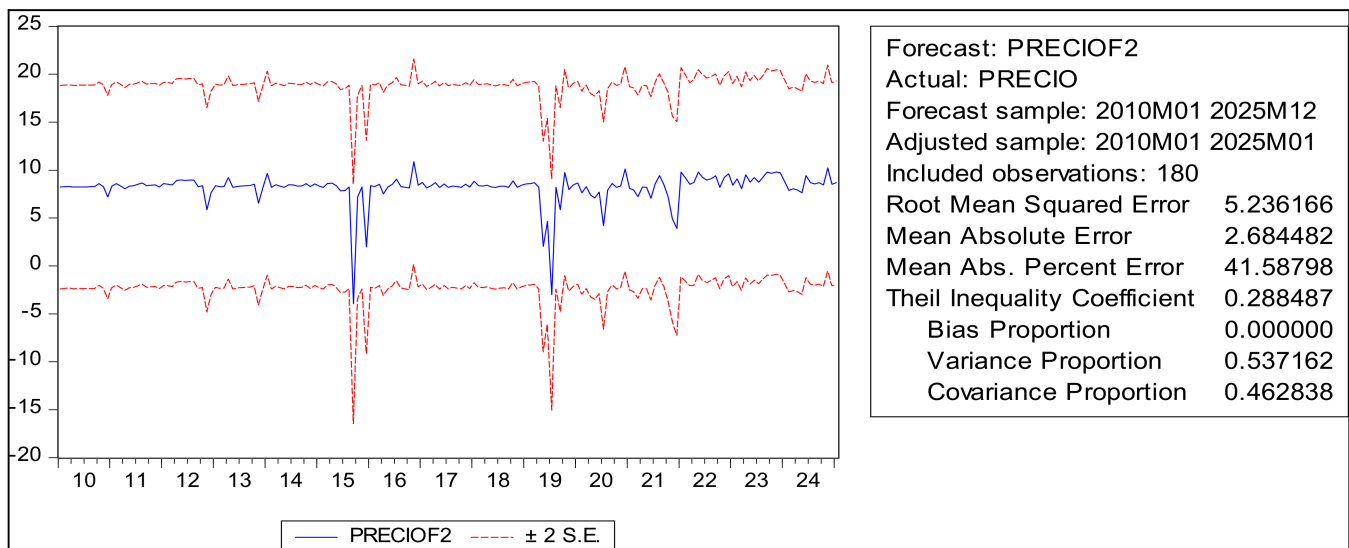
**Table 5.** Causal Forecast Estimate.

| Variable                     | Coefficient           | Std. Error                   | t-Statistic | Prob.    |
|------------------------------|-----------------------|------------------------------|-------------|----------|
| FOB Value                    | $3.67 \times 10^{-5}$ | $1.48 \times 10^{-5}$        | 2.474081    | 0.0143   |
| Net Weight                   | −0.000214             | $6.47 \times 10^{-5}$        | −3.307217   | 0.0011   |
| C                            | 8.211726              | 0.523326                     | 15.69143    | 0.0000   |
| R-squared                    | 0.090661              | Mean dependent variable      |             | 8.154    |
| Adjusted R-squared           | 0.080386              | S.D. of dependent variable   |             | 5.506304 |
| Standard error of regression | 5.280354              | Akaike information criterion |             | 6.182389 |
| Sum of squared residuals     | 4935.138              | Schwarz criterion            |             | 6.235605 |
| Log likelihood               | −553.415              | Hannan–Quinn criterion       |             | 6.203966 |
| F-statistic                  | 8.823413              | Durbin–Watson statistic      |             | 1.821858 |
| Prob (F-statistic)           | 0.000222              |                              |             |          |

Note: Prepared in Eviews 9.0.



**Figure 6.** Residuals from the causal model. Note: Prepared using Eviews 9.0.



**Figure 7.** Causal model forecast. Note: Prepared using Eviews 9.0.

Next, a one-period lagged model was estimated to assess the temporal effect of FOB value and net weight on the Peruvian coffee export price. The results show that none of the lagged variables was statistically significant. Specifically, the one-period lagged FOB value displayed a positive but non-significant coefficient ( $\beta = 1.90 \times 10^{-5}$ ;  $p = 0.221$ ), whereas lagged net weight showed a negative and likewise non-significant coefficient ( $\beta = -1.02 \times 10^{-4}$ ;  $p = 0.134$ ). These findings indicate that, within this specification, the lagged effects of both variables do not significantly explain the dynamics of the export price.

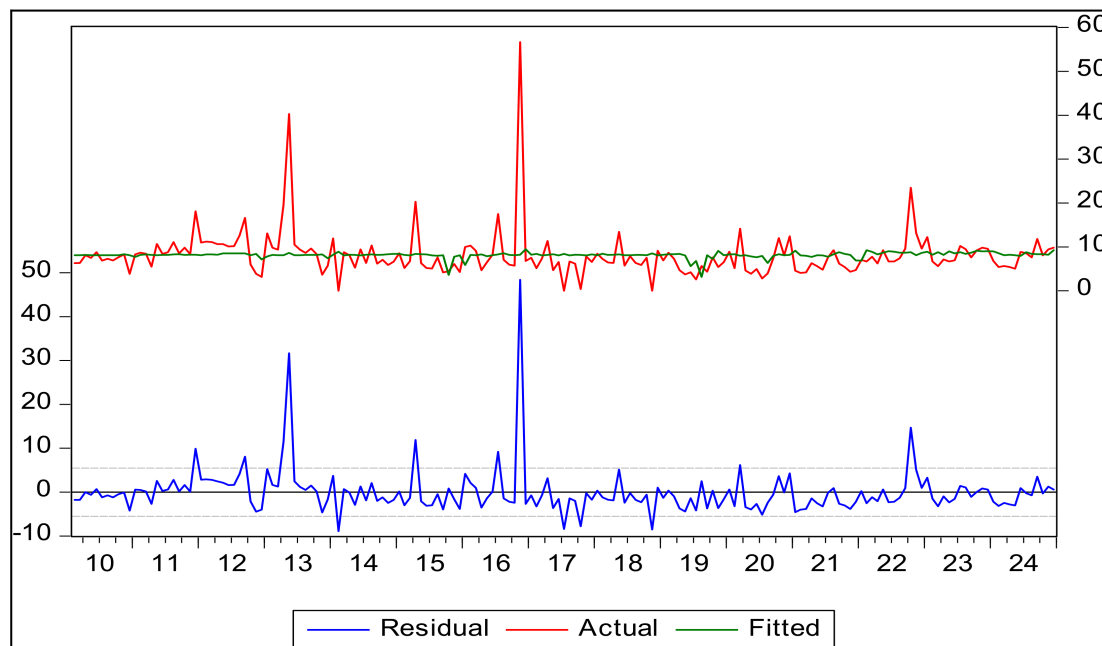
Likewise, the model as a whole was not statistically significant ( $F = 1.459225$ ;  $p = 0.235214$ ), confirming its limited explanatory capacity. This constraint is also reflected in the low coefficient of determination, with an  $R^2$  of 0.016312 and an adjusted  $R^2$  of 0.005133, indicating that only around 1.6% of the variability in the export price is explained by the included variables. In terms of fit, the Akaike Information Criterion was 6.266413, while the Durbin–Watson statistic reached 1.816741, suggesting the possible presence of mild residual autocorrelation. Overall, these results show that the lagged model does not substantially improve the explanation of the Peruvian coffee export price and provides an insufficient fit to describe its behavior, as shown in Table 6 and Figures 8 and 9.

On the other hand, an autoregressive integrated moving average (ARIMA) model was estimated in order to capture the temporal dynamics of the Peruvian coffee export price. The results show that the ARIMA(1,0,0) specification provides the best relative performance among the evaluated models, as it records the lowest Akaike Information Criterion (AIC = 6.221).

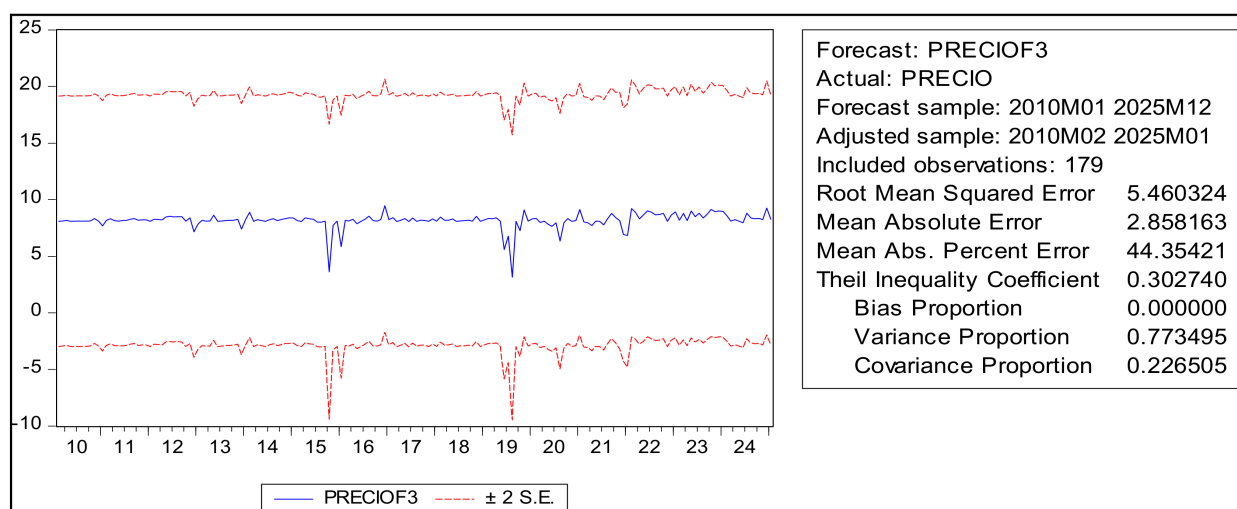
**Table 6.** Estimation of the lag model.

| Variable                     | Coefficient            | Std. Error                   | t-Statistic        | Prob.                 |
|------------------------------|------------------------|------------------------------|--------------------|-----------------------|
| FOB Value (−1)               | $1.90 \times 10^{-5}$  | $1.55 \times 10^{-5}$        | 1.23               | $2.21 \times 10^{-1}$ |
| Net Weight (−1)              | $-1.02 \times 10^{-4}$ | $6.74 \times 10^{-5}$        | −1.51              | $1.34 \times 10^{-1}$ |
| C                            | 8.08                   | $5.47 \times 10^{-1}$        | $1.48 \times 10^1$ | 0.00                  |
| R-squared                    | 0.016312               | Mean of dependent variable   |                    | 8.161397              |
| Adjusted R-squared           | 0.005133               | S.D. of dependent variable   |                    | 5.520852              |
| Standard error of regression | 5.506664               | Akaike information criterion |                    | 6.266413              |
| Sum of squared residuals     | 5336.909               | Schwarz criterion            |                    | 6.319833              |
| Log likelihood               | −557.8439              | Hannan–Quinn criterion       |                    | 6.288074              |
| F-statistic                  | 1.459225               | Durbin–Watson statistic      |                    | 1.816741              |
| Prob (F-statistic)           | 0.235214               |                              |                    |                       |

Note: Prepared using Eviews 9.0.



**Figure 8.** Lag residuals. Note: Prepared using Eviews 9.0 software.



**Figure 9.** Forecast of the lag model. Note: Prepared using Eviews 9.0 software.

In terms of statistical significance, the first-order autoregressive coefficient AR(1) was positive and significant ( $\beta = 0.1393$ ;  $p = 0.011$ ), indicating the presence of temporal dependence in the series; that is, past price values influence its current behavior. In addition, the Durbin–Watson statistic reached 2.004, suggesting the absence of residual autocorrelation and indicating an adequate model fit in terms of serial independence.

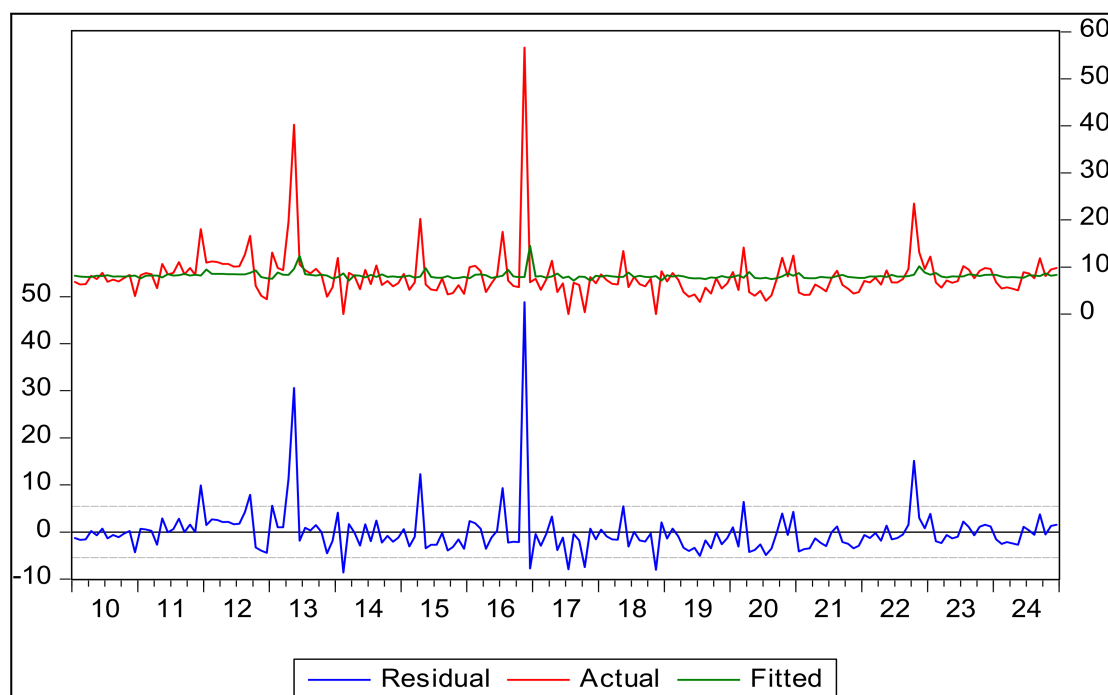
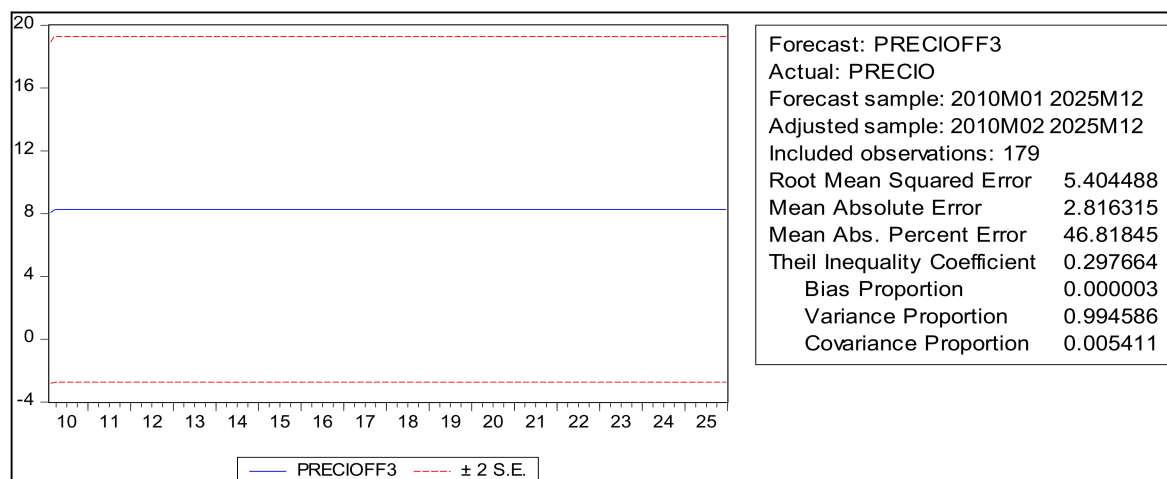
Although the coefficient of determination is relatively low ( $R^2 = 0.0196$ ; adjusted  $R^2 = 0.0085$ ), which is expected in univariate time-series models, the ARIMA specification shows improvements in fit relative to the linear, quadratic, exponential smoothing, and lagged models. Furthermore, the inverted autoregressive root (0.14) lies within the unit circle, confirming the stability of the estimated model.

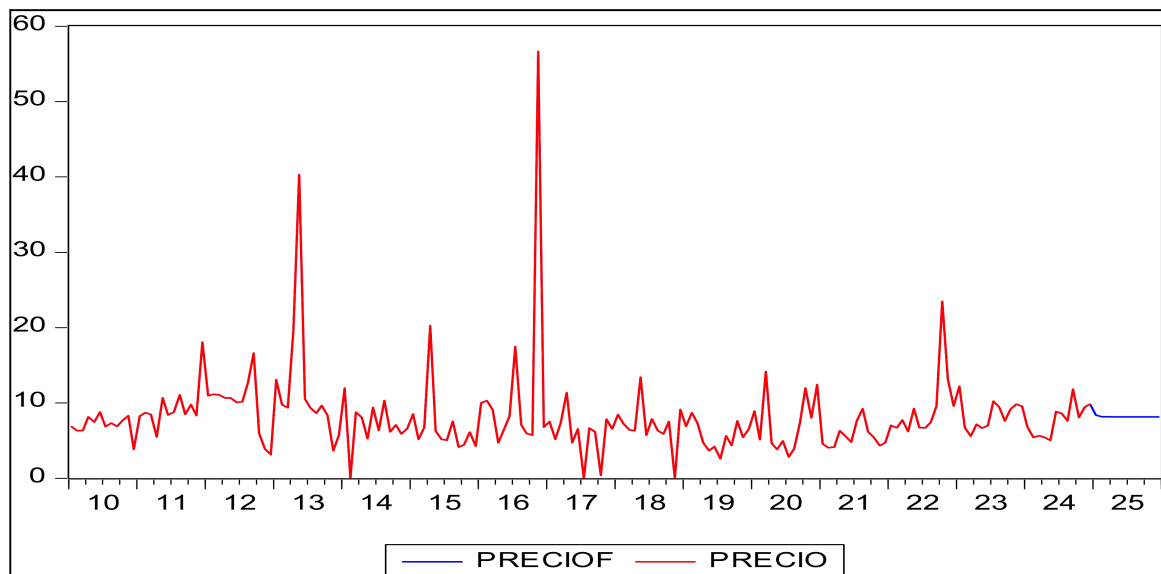
Overall, these results identify the ARIMA (1,0,0) model as the most appropriate specification for describing and forecasting the behavior of the Peruvian coffee export price within the set of evaluated models, as presented in Table 7 and Figures 10–12.

**Table 7.** ARIMA Model Estimation.

| Variable                     | Coefficient | Std. Error                   | t-Statistic | Prob. |
|------------------------------|-------------|------------------------------|-------------|-------|
| C                            | 8.2708      | 0.867981                     | 9.528752    | 0.000 |
| AR(1)                        | 0.1393      | 0.053881                     | 2.584401    | 0.011 |
| SIGMASQ                      | 28.4932     | 1.177188                     | 24.20444    | 0.000 |
| R-squared                    | 0.0196      | Mean dependent variable      |             | 8.271 |
| Adjusted R-squared           | 0.0085      | S.D. of dependent variable   |             | 5.406 |
| Standard error of regression | 5.3829      | Akaike information criterion |             | 6.221 |
| Sum of squared residuals     | 5128.8      | Schwarz criterion            |             | 6.274 |
| Log-likelihood               | −556.89     | Hannan-Quinn criterion       |             | 6.243 |
| F-statistic                  | 1.7683      | Durbin-Watson statistic      |             | 2.004 |
| Prob (F-statistic)           | 0.1736      |                              |             |       |
| Inverted AR Roots            | 0.14        |                              |             |       |

Note: Created in Eviews 9.0

**Figure 10.** ARIMA model residuals. Note: Prepared using Eviews 9.0 software.**Figure 11.** ARIMA model forecast. Note: Prepared using Eviews 9.0 software.



**Figure 12.** ARIMA model forecast, 2010–2024. Note: Prepared using Eviews 9.0.

Finally, the estimation of the GARCH (1,1) model is expressed by the following formula:

$$\text{GARCH} = 2.80573053145 + 0.965869810561 \times \text{RESID}(-1)^2 + 0.462538529459 \times \text{GARCH}(-1),$$

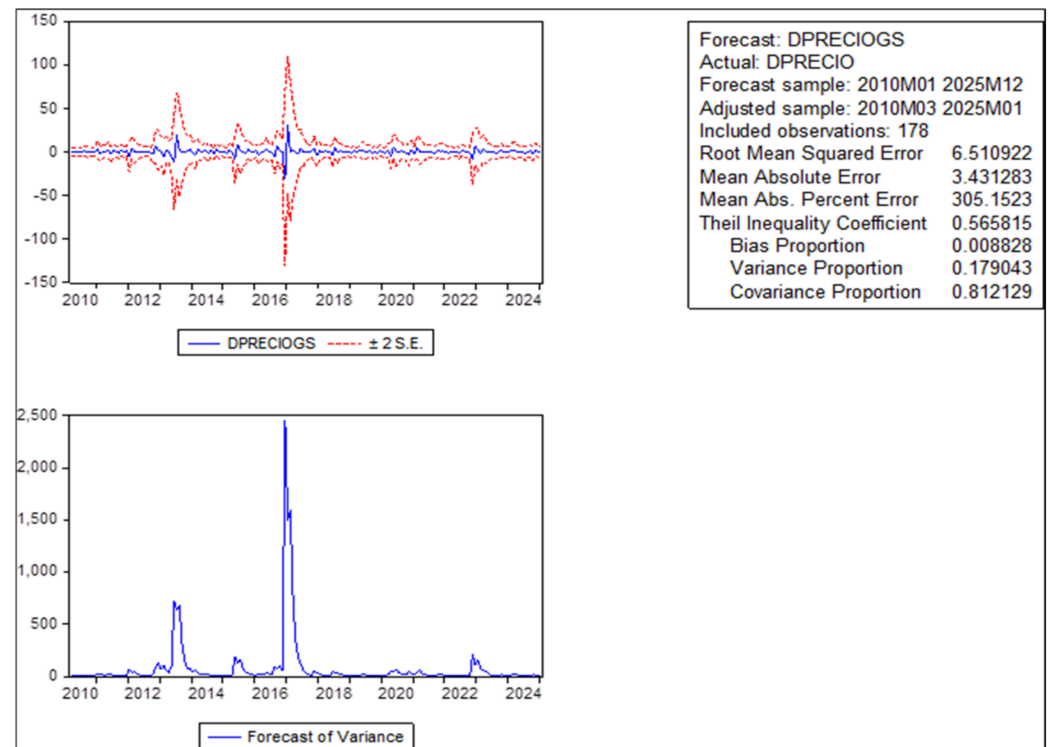
As shown in Table 8, both the AR(1) and GARCH components are significant, confirming the existence of volatility clustering in the Peruvian coffee export price. The conditional variance peaks observed in 2012 and 2016 are linked to shocks associated with the El Niño phenomenon and are interpreted as evidence that past disturbances are the main driver of conditional variance. Likewise, Figure 13 presents the forecast derived from the coffee price estimation.

In addition, a GARCH (1,1) model was estimated because agricultural commodity prices often exhibit episodes of volatility clustering, in which periods of high variability tend to be followed by further periods of high variability. Unlike trend or conditional mean models, the GARCH model explicitly represents time-varying conditional variance, which is appropriate when the analysis is not limited to the expected price level but also considers the magnitude of the risk associated with price fluctuations.

**Table 8.** GARCH Model Estimation.

| Variable               | Coefficient | Std. Error            | z-Statistic | Prob.    |
|------------------------|-------------|-----------------------|-------------|----------|
| C                      | 0.397262    | 0.118988              | 3.338676    | 0.0008   |
| AR(1)                  | −0.612419   | 0.126306              | −4.848681   | 0        |
| Variance Equation      |             |                       |             |          |
| C                      | 2.805731    | 0.666707              | 4.208342    | 0        |
| RESID(−1) <sup>2</sup> | 0.96587     | 0.234675              | 4.115781    | 0        |
| GARCH(−1)              | 0.04625     | 0.075478              | 6.128134    | 0        |
| R-squared              | 0.160979    | Mean dependent var    |             | 0.01676  |
| Adjusted R-squared     | 0.156239    | S.D. dependent var    |             | 7.108391 |
| S.E. of regression     | 6.529516    | Akaike info criterion |             | 6.085314 |
| Sum squared resid      | 7546.32     | Schwarz criterion     |             | 6.174347 |
| Log likelihood         | −539.6356   | Hannan-Quinn criter.  |             | 6.121416 |
| Durbin-Watson stat     | 2.031984    |                       |             |          |
| Inverted AR Roots      | −0.61       |                       |             |          |

Note: Created in Eviews 9.0



**Figure 13.** GARCH model forecast, 2010–2024. Note: Prepared using Eviews 9.0.

### 3.2. Analyzing the Comparison of the Performance of the Different Forecasting Models

Table 9 presents a comparison of the performance of the different forecasting models based on three key criteria: the coefficient of determination ( $R^2$ ), the Akaike Information Criterion (AIC), and the Durbin–Watson statistic. In general, the linear, quadratic, causal, and lagged models exhibit low  $R^2$  values, indicating a limited ability to explain the variability of the coffee export price. In addition, the Durbin–Watson values for these models are below 2, suggesting the presence of residual autocorrelation and potential specification issues that affect the robustness of the estimates.

**Table 9.** Comparative analysis of the models.

| Model      | $R^2$ | Comparison Criteria<br>Akaike Information<br>Criterion | Durbin-<br>Watson | Decision                    |
|------------|-------|--------------------------------------------------------|-------------------|-----------------------------|
| Linear     | 1.1%  | 6.255304                                               | 1.754859          | Presence of autocorrelation |
| Quadratic  | 1.4%  | 6.262977                                               | 1.760906          | Presence of autocorrelation |
| Causal     | 9.1%  | 6.182389                                               | 1.821858          | Presence of autocorrelation |
| Simple lag | 1.6%  | 6.266413                                               | 1.816741          | Presence of autocorrelation |
| ARIMA      | 2.0%  | 6.220984                                               | 2.004359          | No autocorrelation          |
| GARCH      | 16%   | 6.085314                                               | 2.031984          | No autocorrelation          |

Note: Prepared using Eviews 9.0 software.

In contrast, the GARCH model presents the lowest Akaike Information Criterion value (AIC = 6.085314), indicating a better relative fit compared with the other models evaluated. Likewise, the Durbin–Watson statistic is close to 2 (DW = 2.031), suggesting the absence of residual autocorrelation and greater statistical consistency. The heteroskedasticity test shows that Prob. F (1,176) = 0.655 > 0.05; therefore, the model does not present evidence of heteroskedasticity. Although the coefficient of determination of the ARIMA model is relatively low ( $R^2 = 16\%$ ), this result is consistent with the nature of univariate time-series

models, in which predictive performance is not assessed solely on the basis of  $R^2$ , but rather in terms of overall fit and the correct specification of the temporal structure.

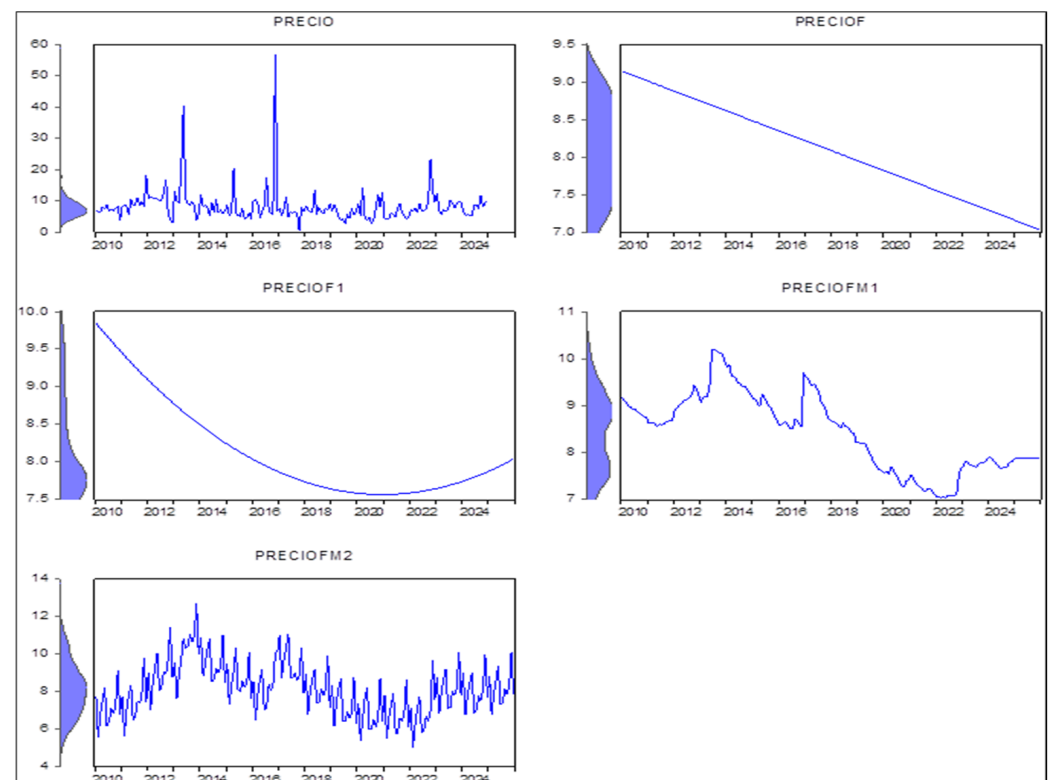
Overall, these results identify the GARCH model as the most appropriate specification for forecasting the Peruvian coffee export price, as it provides a better balance between model fit, parsimony, and reliability in error estimation.

Finally, in line with the general objective of forecasting the behavior of the Peruvian coffee export price in international markets, Table 10 and Figure 14 present the projected values obtained from the different forecasting methodologies for the horizon spanning from December 2024 to December 2025.

**Table 10.** Forecast of coffee price behavior.

| Period  | Linear | Quadratic | Causal | Simple Lag | ARIMA | GARCH |
|---------|--------|-----------|--------|------------|-------|-------|
| 2024M12 | 7.16   | 7.87      | 7.84   | 7.68       | 9.83  | 10.64 |
| 2025M01 | 7.15   | 7.88      | 7.89   | 8.84       | 8.37  | 12.17 |
| 2025M02 | 7.14   | 7.89      | 7.89   | 6.75       | 8.18  | 11.30 |
| 2025M03 | 7.13   | 7.91      | 7.89   | 7.93       | 8.16  | 10.33 |
| 2025M04 | 7.12   | 7.92      | 7.89   | 8.65       | 8.15  | 10.62 |
| 2025M05 | 7.11   | 7.93      | 7.89   | 9.34       | 8.15  | 10.69 |
| 2025M06 | 7.10   | 7.95      | 7.89   | 7.34       | 8.15  | 8.14  |
| 2025M07 | 7.09   | 7.96      | 7.89   | 7.41       | 8.15  | 10.64 |
| 2025M08 | 7.08   | 7.98      | 7.89   | 8.12       | 8.15  | 11.07 |
| 2025M09 | 7.06   | 7.99      | 7.89   | 7.84       | 8.15  | 7.88  |
| 2025M10 | 7.05   | 8.01      | 7.89   | 8.23       | 8.15  | 12.78 |
| 2025M11 | 7.04   | 8.02      | 7.89   | 10.10      | 8.15  | 9.63  |
| 2025M12 | 7.03   | 8.04      | 7.89   | 7.83       | 8.15  | 10.24 |

Note: Prepared using Eviews 9.0 software



**Figure 14.** Forecast of coffee price behavior. Note: Prepared using Eviews 9.0 software.

In general terms, the linear, quadratic, and causal models display relatively similar trajectories, characterized by stable behavior and slight variations toward the end of the

projection period. This pattern suggests structural consistency across these approaches, although with limited capacity to capture more complex changes in price dynamics.

The exponential smoothing model, in turn, shows greater variability in its estimates, with more pronounced fluctuations in certain periods, which may be associated with lower predictive stability in a series with limited seasonal contribution.

Regarding the ARIMA model, it shows more consistent values and a stable trajectory from the first months of 2025 onward. It also makes it possible to model temporal dependence and reduce forecast volatility, thereby providing robust and coherent estimates.

However, considering that the GARCH model is more consistent for projecting coffee prices and shows no evidence of heteroskedasticity or autocorrelation, the comparison with the other models confirms that, although the projected patterns are relatively predictable, the GARCH model offers superior performance in terms of stability and accuracy, consolidating its position as the most appropriate tool for forecasting the Peruvian coffee export price.

#### 4. Discussion

The results of this study indicate that deterministic trend models, both linear and quadratic, fail to adequately explain the dynamics of the Peruvian coffee export price. The lack of statistical significance of their coefficients, together with the presence of residual autocorrelation, suggests that price evolution does not follow simple trend patterns, but rather reflects a more complex temporal structure.

Similarly, the Holt–Winters exponential smoothing model did not demonstrate robust performance. Although it captured some monthly variations, the trend and seasonal components were not statistically relevant, indicating that the series is not primarily driven by a structured seasonal pattern, but rather by persistent internal dynamics.

In the case of the causal model, although FOB value and net export weight were statistically significant, their overall explanatory power remained limited. This finding suggests that export price behavior is not determined solely by contemporaneous observable variables, but also incorporates dynamic and accumulated effects inherent to market processes. Likewise, the lagged model did not improve explanatory capacity, reinforcing the idea that simple exogenous relationships are insufficient to model this series.

In contrast, the GARCH (1,1) model exhibited the best relative performance, as evidenced by the lowest AIC value, the statistical significance of the autoregressive component, and the absence of residual autocorrelation. These results confirm the presence of temporal dependence, indicating that current price behavior is influenced by past values. This finding is consistent with the nature of commodity markets, where price formation incorporates expectations, shock transmission, and gradual adjustments.

These findings are consistent with previous literature. Ref. [39] highlights the strong performance of ARIMA models in in-sample forecasting, while Ref. [44] emphasizes their usefulness in coffee price analysis, particularly when combined with volatility models such as GARCH. Similarly, Ref. [67] demonstrates their applicability in agricultural export markets. However, other studies, such as Ref. [64] and Ref. [45], show that nonlinear and hybrid approaches may outperform traditional models when more complex structures are present, suggesting that model selection depends on the specific characteristics of the data.

From an economic perspective, the results confirm that coffee export prices are not solely determined by the contemporaneous interaction of supply and demand, but also by mechanisms of temporal persistence associated with market conditions, climatic shocks, logistics constraints, and international regulations. This is particularly relevant in the Peruvian context, where price volatility and production constraints affect income stability.

Regarding the projections, the GARCH model shows a volatile trajectory, with values converging around 10 from March 2025 onward. This implies that the coffee price is highly volatile and that relevant aspects such as export planning, contract negotiation, and risk management must be considered. Accordingly, these results suggest that the superiority of the GARCH model lies in its ability to dynamically represent the temporal structure of the series in a coherent manner and to improve predictive accuracy. However, these results should be interpreted with caution, since the analysis is based on a univariate framework, although other variables that influence the model could be incorporated to improve the prediction of the coffee market price.

## 5. Conclusions

This study estimated and compared seven econometric and time-series models to forecast the behavior of Peruvian coffee export prices in international markets for 2025. The findings showed that the GARCH (1,1) model achieved the best performance among the specifications tested, as it reported the lowest Akaike information criterion ( $AIC = 6.085$ ), a Durbin–Watson statistic close to 2.00 ( $DW = 2.032$ ), an  $R^2$  of 16%, and no ARCH effects in the residuals. The 2025 projections indicate a fluctuating trajectory that is expected to remain close to 10.00 from March onward. These results confirm that uncertainty persists in the Peruvian coffee export market and that price forecasting requires models capable of capturing volatility dynamics.

The different models confirmed differences in the adjustment of the export price and in the way the series was represented. The deterministic models, both linear and quadratic, did not adequately reflect the behavior of the series due to the lack of statistical significance in their trend components and the presence of residual autocorrelation. Similarly, the Holt–Winters model did not provide a robust representation of the series, since neither trend nor seasonality contributed significantly to the model structure. Although the causal model identified statistically significant relationships among FOB value, net weight, and the export price, its explanatory power remained limited ( $R^2 = 9.1\%$ ), whereas the lagged model did not yield statistically relevant results. The ARIMA(1,0,0) model showed adequate temporal dependence, absence of autocorrelation ( $DW = 2.004$ ), and the second-lowest AIC (6.221), confirming it as a solid univariate benchmark. However, its limited ability to model conditional variance placed it below the GARCH (1,1) specification in terms of overall predictive performance.

Consequently, the comparative test showed that forecasting the Peruvian coffee export price provides better results when the model incorporates changes in volatility over time rather than only average price behavior. The significance of the ARCH and GARCH components demonstrated the presence of volatility clustering in the series, a pattern commonly observed in agricultural commodity markets affected by climatic, commercial, and regulatory changes. As a result, the GARCH (1,1) model constitutes a reliable instrument to support decision-making in the coffee export sector. This is particularly useful for export companies, producer organizations, cooperatives, PROMPERÚ, chambers of commerce, and foreign trade institutions involved in market intelligence, export promotion, and commercial risk management. Its use can strengthen export planning, contract negotiation, price monitoring, inventory and shipment scheduling, and the design of evidence-based market strategies. Future studies should validate these projections using actual 2025 prices and incorporate climatic, logistical, financial, and regulatory factors to strengthen export price forecasting.

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